Thesis Methodology Terminologies & Mathematical Reference Guide

Age-Stratified Dengue Severity Prediction and Subgroup-Conditioned Feature Attribution Analysis

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> Purpose: Exhaustive reference manual for thesis defense covering every methodological concept, working flow, equation, and clinical interpretation.

1. Dataset Preprocessing & Cohort Partitioning

- **Dengue NS1 Antigen (Non-Structural Protein 1):** A highly conserved glycoprotein secreted by the dengue virus into human serum during the acute febrile/viremic phase (Days 1–7).
  - **Working Procedure:** Raw dataset ingestion (1,329 records) -> Filtering records -> Retaining only confirmed NS1-positive cases (972 records).
  - **Role in Thesis:** Guarantees that models are trained strictly on biologically validated active dengue cases, removing non-dengue febrile confounders.

- **Cohort Partitioning (Age Stratification):** Subdividing the global population into clinically distinct age brackets to uncover subgroup-specific predictive patterns.
  - **Working Procedure:** Divided 972 NS1-positive records into Pediatric (Age <= 18, n=158) and Adult (Age > 18, n=814), keeping an Age-Pooled cohort (n=972) as baseline.
  - **Role in Thesis:** Children exhibit higher microvascular fragility and rapid capillary leakage compared to adults; stratification prevents adult majority dominance.
- **Median Imputation:** A non-parametric missing-data replacement method where missing continuous values are substituted by the middle value of sorted observed features.
  - **Working Procedure:** 1. Extract non-null values for feature X_j: $x_{(1)} \le x_{(2)} \le \dots \le x_{(m)}$. 2. Compute median $\tilde{x}_j$. 3. Substitute all missing cells with $\tilde{x}_j$.
  - **Role in Thesis:** Clinical enzymes like AST and ALT exhibit heavy right-skewed tails from acute liver injury; median is robust against extreme outlier distortion.

$$\tilde{x}_j = x_{((m+1)/2)} \quad [\text{if } m \text{ is odd}] \quad \text{OR} \quad 0.5 * \left(x_{(m/2)} + x_{(m/2+1)} \right) \quad [\text{if } m \text{ is even}]$$

Where: m = number of non-missing samples for feature j

- **Routine Hematology Predictors:** Standard Complete Blood Count (CBC) and liver function markers routinely available in resource-limited emergency triage settings.

- **Working Procedure:** Includes White Blood Cell (WBC) count, Hematocrit (HCT), Lymphocyte %, Neutrophil %, AST (SGOT), and ALT (SGPT).
- **Role in Thesis:** Enables early, low-cost risk scoring before severe shock occurs, without requiring expensive specialized molecular assays.

2. Target Proxy Harmonization & Target Leakage Prevention

- **Target Proxy Harmonization:** Defining a clinically grounded binary surrogate label when official WHO 2009 multi-organ composite scores are unavailable in retrospective datasets.
  - **Working Procedure:** Following Niazi & Momand (2026), platelet count (PLT) was stratified: Severe (< 50,000 / µL), Moderate (50,000 - 100,000 / µL), Minor (> 100,000 / µL). Moderate and Minor were combined into Non-Severe.
  - **Role in Thesis:** Produces a clear binary outcome (Severe vs Non-Severe) matching clinical urgency for platelet transfusion/ICU admission.

$$y_i = 1 \text{ (Severe) if } PLT_i < 50{,}000 \ / \ \mu L \ \ | \ \ y_i = 0 \text{ (Non-Severe) if } PLT_i >= 50{,}000 \ / \ \mu L$$

Where: y_i = binary outcome variable for patient i

- **Target Leakage Prevention:** The systematic removal of features that directly contain information about the target variable, preventing trivial, over-optimistic performance.

- **Working Procedure:** Because y is derived directly from PLT, Platelet Count (PLT) and all its mathematical ratios (e.g., PLT/WBC) were strictly excluded from feature matrix X prior to model training.
- **Role in Thesis:** Forces the ML models to identify true pathophysiological antecedents (leukopenia, transaminases, hemoconcentration) rather than trivial identity mappings.

3. Clinical Feature Engineering

Domain-engineered binary indicators and interaction ratios based on standard clinical hematology guidelines:

Feature Name	Mathematical Logic / Threshold	Clinical Significance
Leukopenia Flag	I(WBC < 4,000 cells/mm^3)	Bone marrow suppression & virus-induced bone marrow hypoplasia in early dengue.
Lymphopenia Flag	I(Lymphocyte % < 20%)	Acute depletion of peripheral T-cells and circulating lymphocytes during early viremia.
Neutrophilia Flag	I(Neutrophil % > 70%)	Relative neutrophil dominance under stress response or secondary bacterial overlap.
Transaminase Ratio (De Ritis Ratio)	Ratio = AST / ALT	Reflects disproportionate AST elevation characteristic of dengue hepatic injury.

Elevated Transaminases Flag	I(AST > 40 U/L or ALT > 40 U/L)	Direct clinical indicator of acute dengue hepatitis / hepatic parenchymal damage.

4. Dataset Splitting & Imbalance Management

- **Stratified Train-Test Split (80/20):** A partitioning strategy where the 80% training set and 20% test set maintain the exact same ratio of Severe vs Non-Severe cases as the original dataset.
  - **Working Procedure:** Implemented using scikit-learn's StratifiedShuffleSplit (random_state = 42).
  - **Role in Thesis:** Prevents severe cases from being disproportionately assigned to either train or test splits, ensuring unbiased evaluation.
- **Balanced Class Weighting:** A cost-sensitive training technique that penalizes misclassifications in the minority class (Severe) more heavily than in the majority class.
  - **Working Procedure:** Adjusts class weights in model loss functions inversely proportional to class frequencies.
  - **Role in Thesis:** Avoids model collapse where a classifier blindly predicts the majority non-severe class to achieve deceptively high raw accuracy.

$$w\_c = N / (K * n\_c)$$

Where: w_c = weight for class c; N = total samples; K = total classes (2); n_c = samples in class c

5. Machine Learning Classifiers & Ensemble Architectures

- **Logistic Regression (LR):** A generalized linear classifier that models the log-odds of severe dengue as a linear combination of hematological features via the sigmoid function.
  - **Working Procedure:** Trained with L2 ridge regularization and balanced class weights as a transparent linear baseline.
  - **Role in Thesis:** Provides a baseline comparison against non-linear tree-based ensembles.

$$P(y=1 \mid x) = \mathrm{sigma}(w^T x + b) = 1 / (1 + \exp(-(w^T x + b)))$$

Where: w = feature weight vector, b = bias intercept, x = input feature vector

- **Random Forest (RF) (Breiman, 2001):** An ensemble of de-correlated decision trees constructed via Bootstrap Aggregation (Bagging) and Random Feature Subspace Selection.
  - **Working Procedure:** 1. Sample N observations with replacement. 2. At each node, select m = sqrt(d) random features. 3. Split by minimizing Gini impurity. 4. Average votes across all trees.
  - **Role in Thesis:** Highest performing model for the Adult cohort (Accuracy = 84.7%, AUC = 0.892, F1 = 0.820).

$$I_G(t) = 1 - \mathrm{sum}(p(k|t)^2) = 1 - (p_0^2 + p_1^2)$$

Where: I_G(t) = Gini impurity at tree node t; p(k|t) = proportion of class k at node t

- **XGBoost (Extreme Gradient Boosting) (Chen & Guestrin, 2016):** A highly regularized sequential boosting algorithm that fits successive decision trees to the negative gradient of the loss function using second-order Taylor expansion.
  - **Working Procedure:** Computes exact 1st (gradient g_i) and 2nd (hessian h_i) order derivatives to evaluate split quality while penalizing tree complexity.
  - **Role in Thesis:** Primary model utilized for computing TreeSHAP feature attributions and dependence curves.

$$Obj^{(t)} = \sum \left[g_i * f_t(x_i) + 0.5 * h_i * (f_t(x_i))^2 \right] + \gamma * T + 0.5 * \lambda * \sum (w_j^2)$$

Where: g_i = gradient; h_i = hessian; T = leaf count; w_j = leaf weights; gamma, lambda = regularization terms

- **LightGBM (Light Gradient Boosting Machine) (Ke et al., 2017):** A high-speed GBDT variant utilizing Gradient-based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB) with leaf-wise (best-first) tree growth.
  - **Working Procedure:** Expands the leaf with the highest loss reduction rather than splitting level-wise across the entire tree depth.
  - **Role in Thesis:** Provides ultra-fast training and handles potential non-linear feature interactions efficiently.
- **Stacking Ensemble Meta-Learner (Wolpert, 1992):** A heterogeneous ensemble technique where predictions of multiple Level-0 base classifiers serve as input meta-features for a Level-1 meta-classifier.

- **Working Procedure:** Level-0 models (LR, RF, XGBoost, LightGBM) generate out-of-fold probability vectors $z_i = [p_{LR}, p_{RF}, p_{XGB}, p_{LGBM}]$, which a Level-1 Logistic Regression meta-learner combines.
- **Role in Thesis:** Highest performing model for the Pediatric cohort (Accuracy = 90.6%, AUC = 0.947, F1 = 0.800, Specificity = 100%).

6. Model Evaluation Metrics & Formulations

Let TP = True Positives (Severe), TN = True Negatives (Non-Severe), FP = False Positives, FN = False Negatives.

Metric Name	Mathematical Equation	Clinical Interpretation
Accuracy (Acc)	Acc = (TP + TN) / (TP + TN + FP + FN)	Overall proportion of correct classifications across all patients.
Sensitivity / Recall (TPR)	Sens = TP / (TP + FN)	Ability to catch true severe cases. High sensitivity is crucial to prevent missed fatal emergencies.
Specificity (TNR)	Spec = TN / (TN + FP)	Ability to correctly identify non-severe patients, preventing unnecessary hospital overcrowding.
F1-Score	F1 = 2 * (Prec * Rec) / (Prec + Rec) = 2*TP / (2*TP + FP + FN)	Harmonic mean of precision and recall; robust measure for imbalanced medical data.

AUC-ROC	AUC = Integral(TPR(FPR^-1(t)) dt)	Probability that the model ranks a randomly chosen severe patient above a non-severe patient.
Matthews Correlation Coefficient (MCC)	MCC = (TP*TN - FP*FN) / sqrt((TP+FP)(TP+FN)(TN+FP)(TN+FN))	Balanced correlation coefficient ranging from -1 to +1; unaffected by class imbalance.

7. Explainable AI (XAI) & SHAP Analysis

- **SHapley Additive exPlanations (SHAP) (Lundberg & Lee, 2017):** A game-theoretic framework that computes the unique, fair additive marginal contribution (phi_i) of each clinical feature towards the model's severity prediction.
  - **Working Procedure:** Calculates the weighted average marginal contribution of feature i across all possible feature subsets (coalitions) S subset of F \ {i}.
  - **Role in Thesis:** Provides mathematically unified local (patient-specific) and global (cohort-wide) model explanations.

> phi_i(f, x) = sum_(S subset F \ {i}) [|S|! * (|F| - |S| - 1)! / |F|!] * [f_x(S union {i}) - f_x(S)]
>
> *Where: |F| = total features; S = feature coalition excluding i; f_x(S union {i}) - f_x(S) = marginal gain of adding feature i*

- **Four Fundamental Axioms of SHAP:** Guarantees that feature attributions are theoretically sound and consistent:

    - **Working Procedure:** 1. Efficiency (Local Accuracy): sum(phi_i) = f(x) - E[f(X)]. 2. Symmetry: identical contributors receive identical phi. 3. Dummy Player: non-contributing features receive phi = 0. 4. Additivity: phi of ensemble sum equals sum of component phi values.
    - **Role in Thesis:** Ensures attributions cannot be manipulated or produce contradictory feature rankings.
- **TreeExplainer (TreeSHAP):** A specialized, polynomial-time algorithm O(T * L * D^2) for calculating exact Shapley values on tree-based ensembles without exponential brute-force sampling.
    - **Working Procedure:** Recursively traverses tree structures of trained XGBoost models to compute conditional expectations.
    - **Role in Thesis:** Allows rapid, exact explanation computation on all 972 patient records.
- **SHAP Summary (Beeswarm) & Dependence Plots:** Visual analytics displaying feature importance ranking, directionality of impact, and non-linear risk trajectories.
    - **Working Procedure:** Beeswarm plot plots every patient as a dot along the SHAP axis (Red = High feature value, Blue = Low feature value). Dependence plot graphs exact feature value (x-axis) vs SHAP risk (y-axis).
    - **Role in Thesis:** Uncovered that elevated Lymphocyte % drives severe risk in both cohorts, but with starkly different non-linear inflection points.

8. Subgroup Divergence & Programmatic Cutoff Extraction

- **Kendall's Tau-b (tau_b) Rank Correlation:** A non-parametric statistical coefficient measuring the relative ordering concordance between pediatric and adult SHAP feature rankings while adjusting for ties.
  - **Working Procedure:** Calculated by comparing concordant pairs P and discordant pairs Q between the two cohort feature importance lists.
  - **Role in Thesis:** Resulted in tau_b = 0.900, proving high overall relative rank agreement (both cohorts prioritize Lymphocytes, WBC, and AST), while threshold values diverge.

$$\text{tau\_b} = (P - Q) / \text{sqrt}((P + Q + T\_x) * (P + Q + T\_y))$$

Where: P = concordant pairs; Q = discordant pairs; T_x, T_y = ties in ranking x and y

- **Programmatic Cutoff Extraction via Depth-1 Decision Trees (Decision Stumps):** A machine learning post-processing method that fits single-split decision trees to continuous SHAP values to extract exact, actionable clinical cutoff thresholds.
  - **Working Procedure:** 1. Define binary risk target z_i = I(SHAP_feature(x_i) > 0). 2. Train a Decision Tree Classifier (max_depth = 1) on feature x. 3. Extract the root node split point theta*.
  - **Role in Thesis:** Translates continuous, abstract XAI attribution curves into concrete clinical triage rules.

$$\text{theta}^* = \text{argmin\_theta} \, [\, (N\_L / N) * \text{Impurity}(D\_L(\text{theta})) + (N\_R / N) * \text{Impurity}(D\_R(\text{theta})) \,]$$

> *Where: theta* = optimal split cutoff; D_L = {x_i <= theta}; D_R = {x_i > theta}*

Empirically Extracted High-Risk Decision Cutoffs (SHAP > 0)

Clinical Predictor	Pediatric High-Risk Split (<= 18 yrs)	Adult High-Risk Split (> 18 yrs)
Lymphocyte Percentage	> 44.50%	> 24.99%
White Blood Cell (WBC) Count	<= 4.34 x 10^3 cells/mm^3	Linear negative risk trend
Neutrophil % / Hematocrit	Neutrophil % <= 48.00%	Hematocrit (HCT) > 40.65% (Age > 31.00 yrs)

> 📌 **CORE THESIS DEFENSE QUESTIONS & ANSWERS:** Q1: Why exclude PLT from training? -> To prevent target leakage since severe status was derived from PLT < 50,000/µL.
>
> Q2: Why Kendall's tau-b = 0.900? -> Shows top features are identical in rank, but quantitative decision boundaries differ markedly between cohorts.
>
> Q3: Why did Stacking win on Pediatrics (90.6%) and RF on Adults (84.7%)? -> Stacking meta-learning controlled high variance in the smaller pediatric cohort (n=158), while RF effectively handled non-linear boundaries in the larger adult cohort (n=814).
>
> Q4: Why depth-1 decision trees on SHAP? -> Bridges the gap between black-box AI and clinical medicine by transforming

> continuous attribution curves into testable numerical threshold rules.